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1 Main Figures

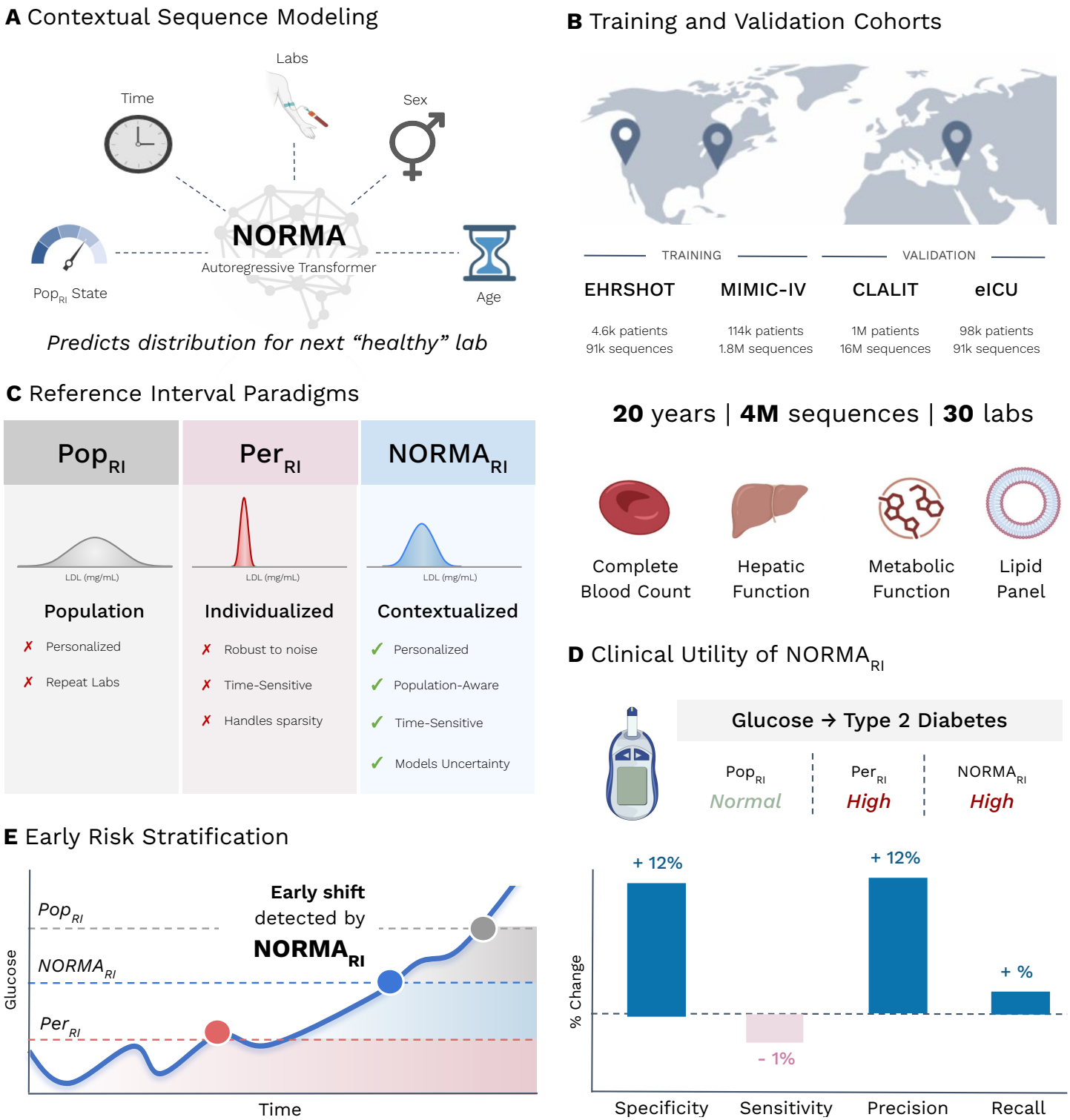
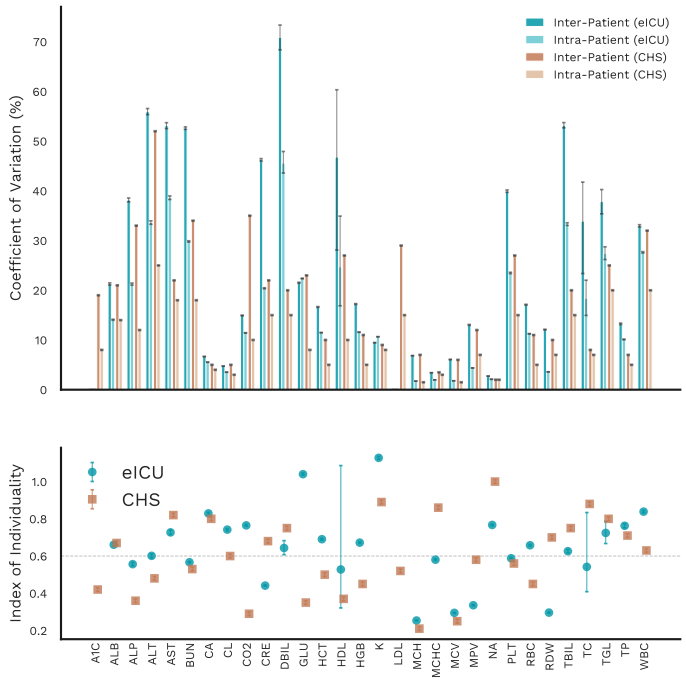


Figure 1: Overview of NORMA. (A) Contextual sequence modeling: NORMA is an autoregressive transformer that integrates labs, time, sex, age, and population-level state to predict the distribution of the next healthy lab value. (B) Training and validation cohorts spanning 20 years, 4M sequences, and 30 analytes across four datasets. (C) Reference interval paradigms: population (PopRI), individualized (PerRI), and contextualized (NORMA_{RI}). (D) Clinical utility of NORMA_{RI} for outcome prediction. (E) Early risk stratification via detection of shifts before population thresholds.

A Cohort Selection

Insert cohort selection schematic

B Intra-Patient vs Inter-Patient Variation



C Mortality Association

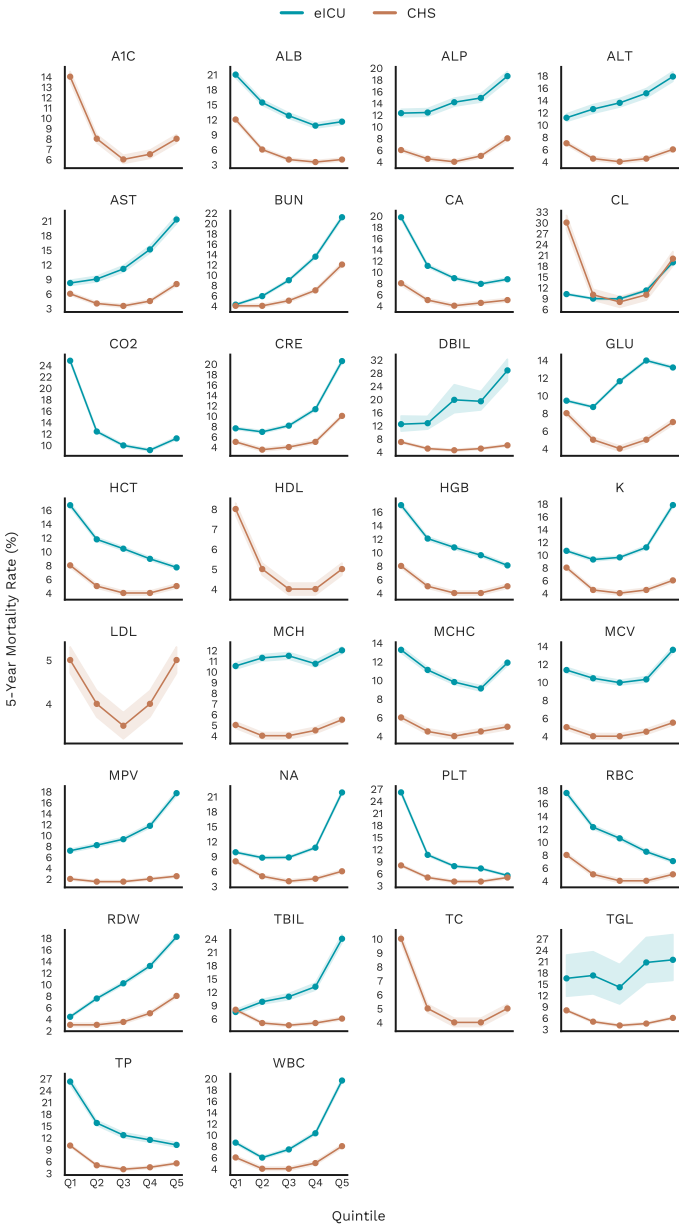


Figure 2: Detection of within-person abnormalities by personalized intervals. (A) Cohort selection schematic. (B) Intra-patient vs inter-patient coefficient of variation and index of individuality across 30 analytes. (C) Five-year mortality rate by quintile of raw analyte value for eICU and CHS cohorts.

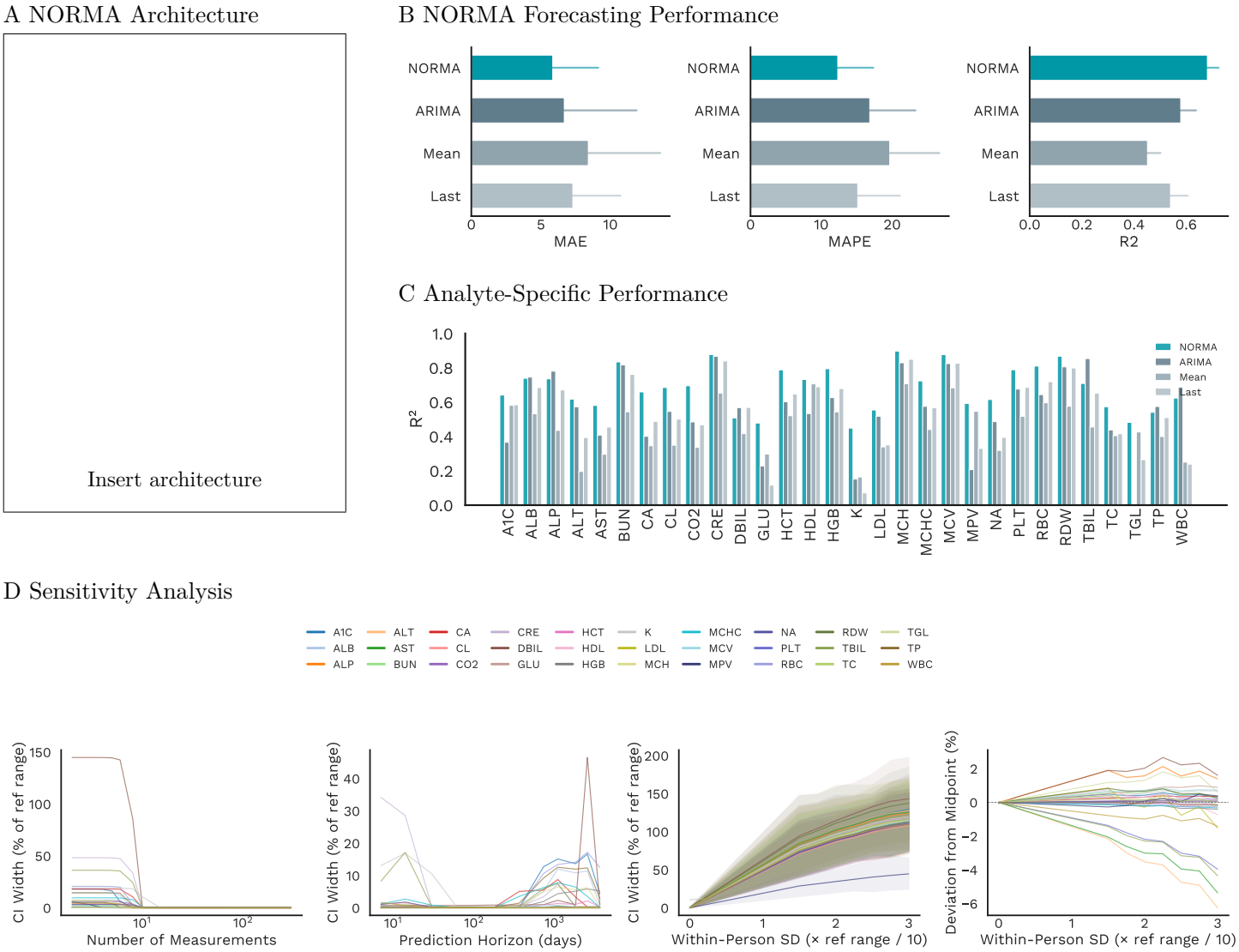


Figure 3: NORMA architecture and predictive performance. (A) Model architecture: decoder-only transformer with context, history, and query tokens producing quantile predictions. (B) Forecasting accuracy (MAE, MAPE, R^2) compared with ARIMA, patient mean, and last-value baselines. (C) Per-analyte R^2 across 30 laboratory tests. (D) Sensitivity analysis: effect of history length, prediction horizon, and within-person variability on prediction interval width.

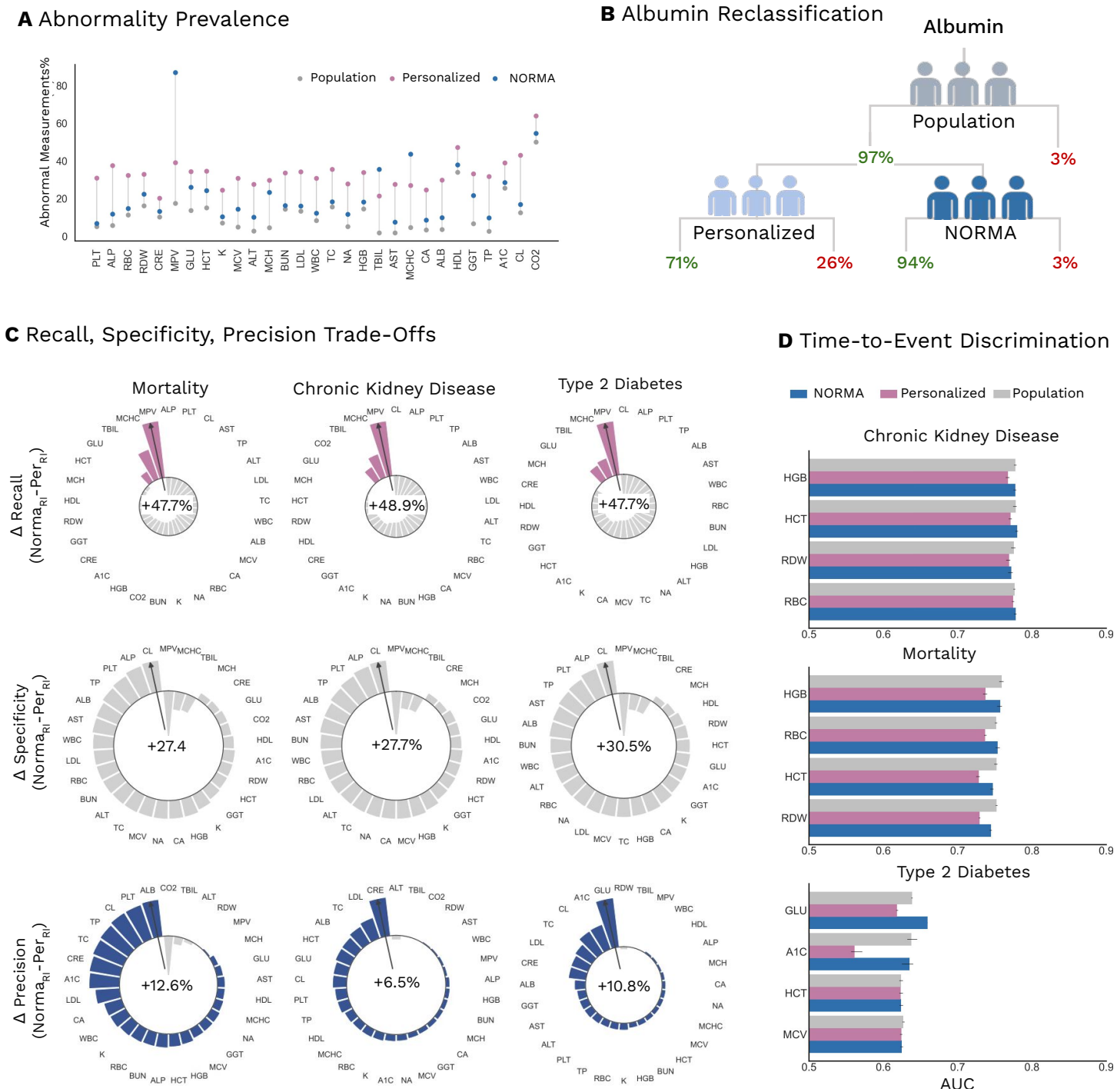


Figure 4: Time-dependent discrimination for clinical outcomes in Clalit Health Services. (A) Abnormality prevalence across reference interval frameworks (PopRI, PerRI, NORMA_{RI}). (B) Albumin reclassification across paradigms. (C) Recall, specificity, and precision trade-offs (circos plots) for mortality, CKD, and type 2 diabetes. (D) Time-to-event discrimination (AUC) by analyte and method.

Figure 5: Clinical outcome prediction in eICU. (A) Reclassification rate among PopRI-normal measurements by PerRI and NORMA_{RI}. (B) Cox proportional hazard ratios for mortality (significant associations only). (C) Sensitivity vs false positive rate for acute kidney injury, mortality, prolonged ICU stay, and sepsis. (D) Lead time: how much earlier NORMA flags abnormalities compared with PopRI. (E) Recall, specificity, precision, and accuracy trade-offs by outcome.

2 Supplemental Figures

	History						Query
dt (days)	0	90	240	380	387	2	1
Value	8.9	9.2	10.5	12.6	13.1	13.4	X_t
Outpatient	1	1	0	1	1	1	1

Figure S1: Supplementary Figure 1. Example token sequence input to NORMA. Each measurement is represented by a time gap (days since previous), the lab value, and a binary outpatient indicator. The model predicts the distribution of the next value X_t.

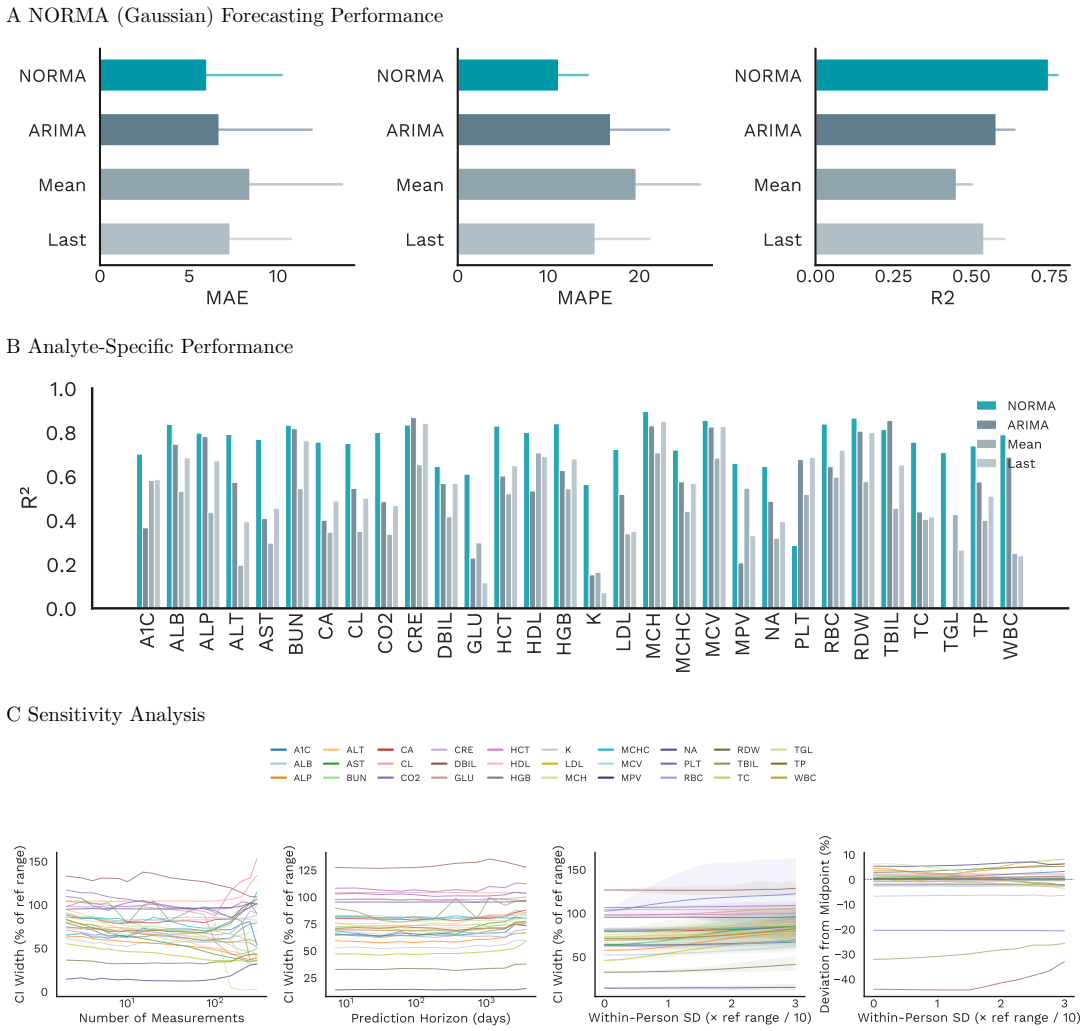


Figure S2: Supplementary Figure 2. NORMA (Gaussian) predictive performance. (A) Forecasting accuracy (MAE, MAPE, R^2) compared with baselines. (B) Per-analyte R^2. (C) Sensitivity analysis: effect of history length, prediction horizon, and within-person variability.

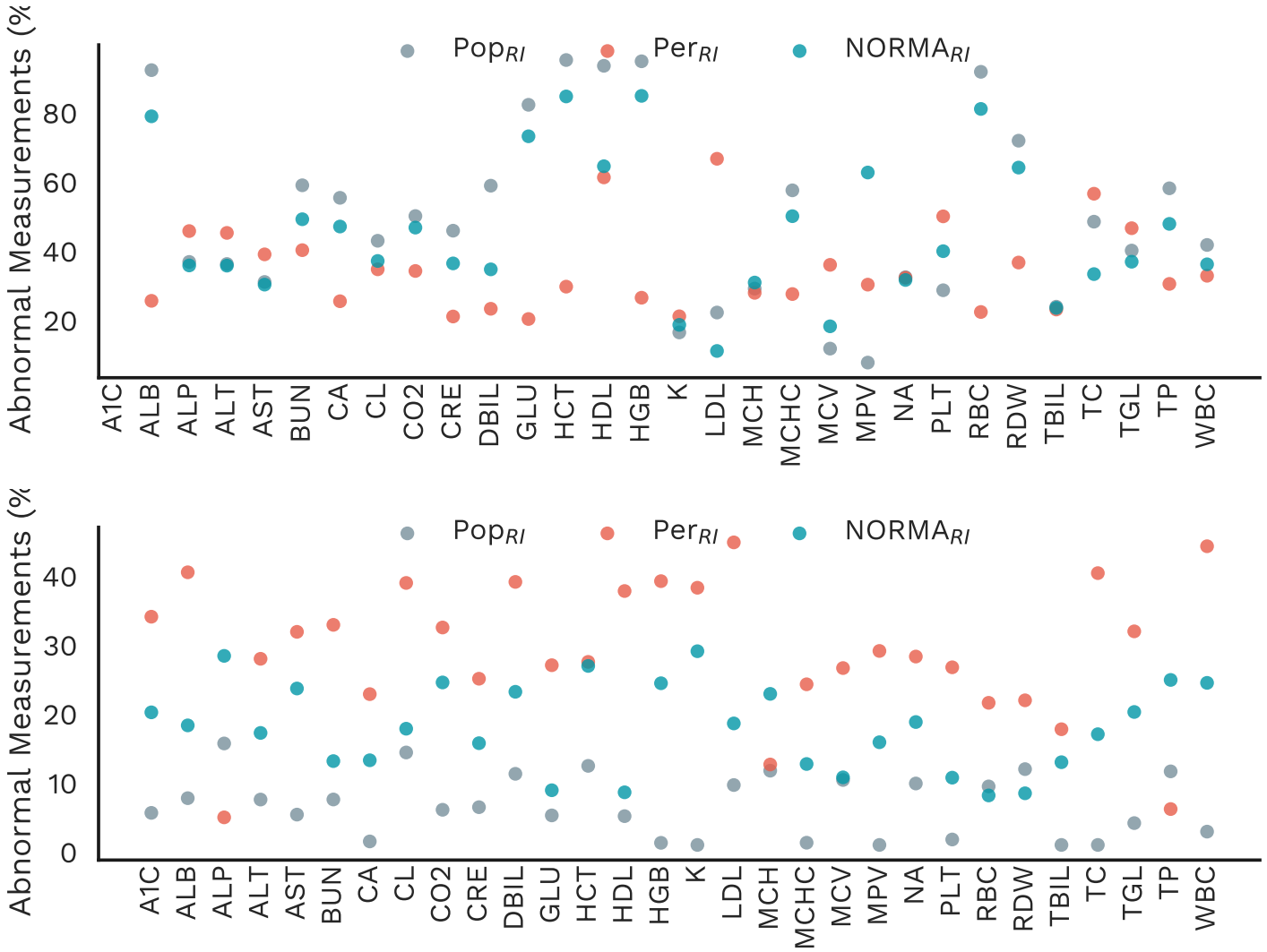


Figure S3: Supplementary Figure 3. Abnormality prevalence across reference interval frameworks. Top: eICU. Bottom: CHS.

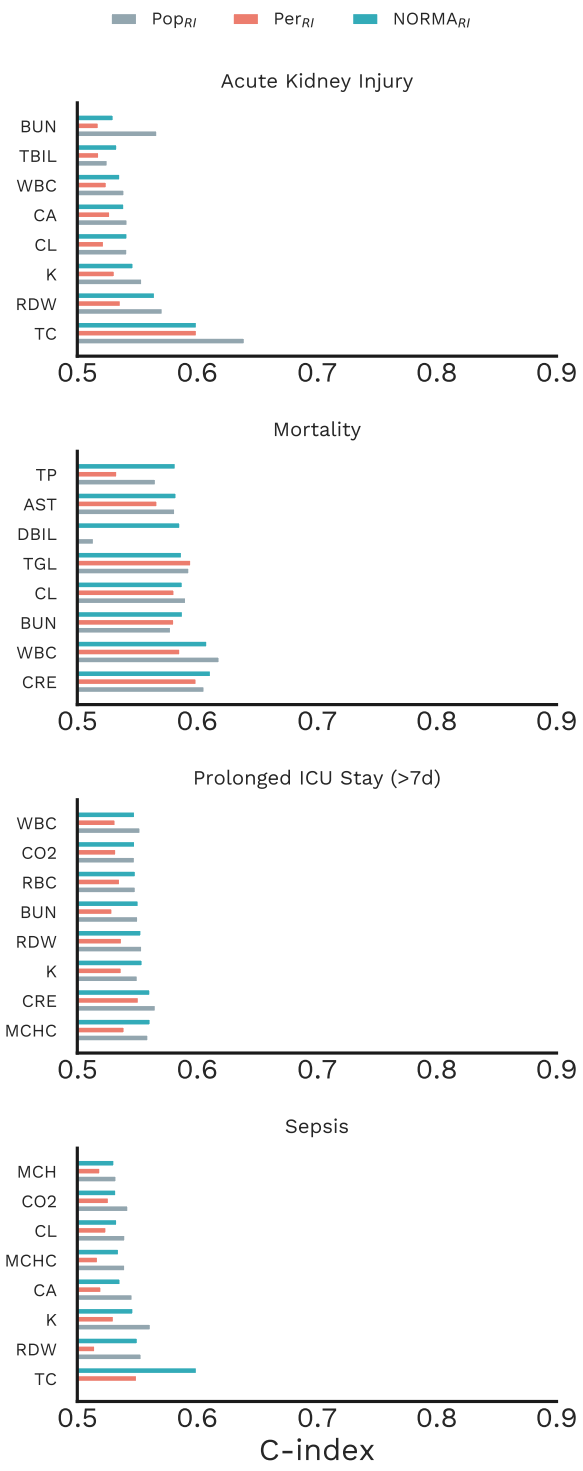


Figure S4: Supplementary Figure 4. Concordance index for Cox time-to-event models by analyte and method in eICU.

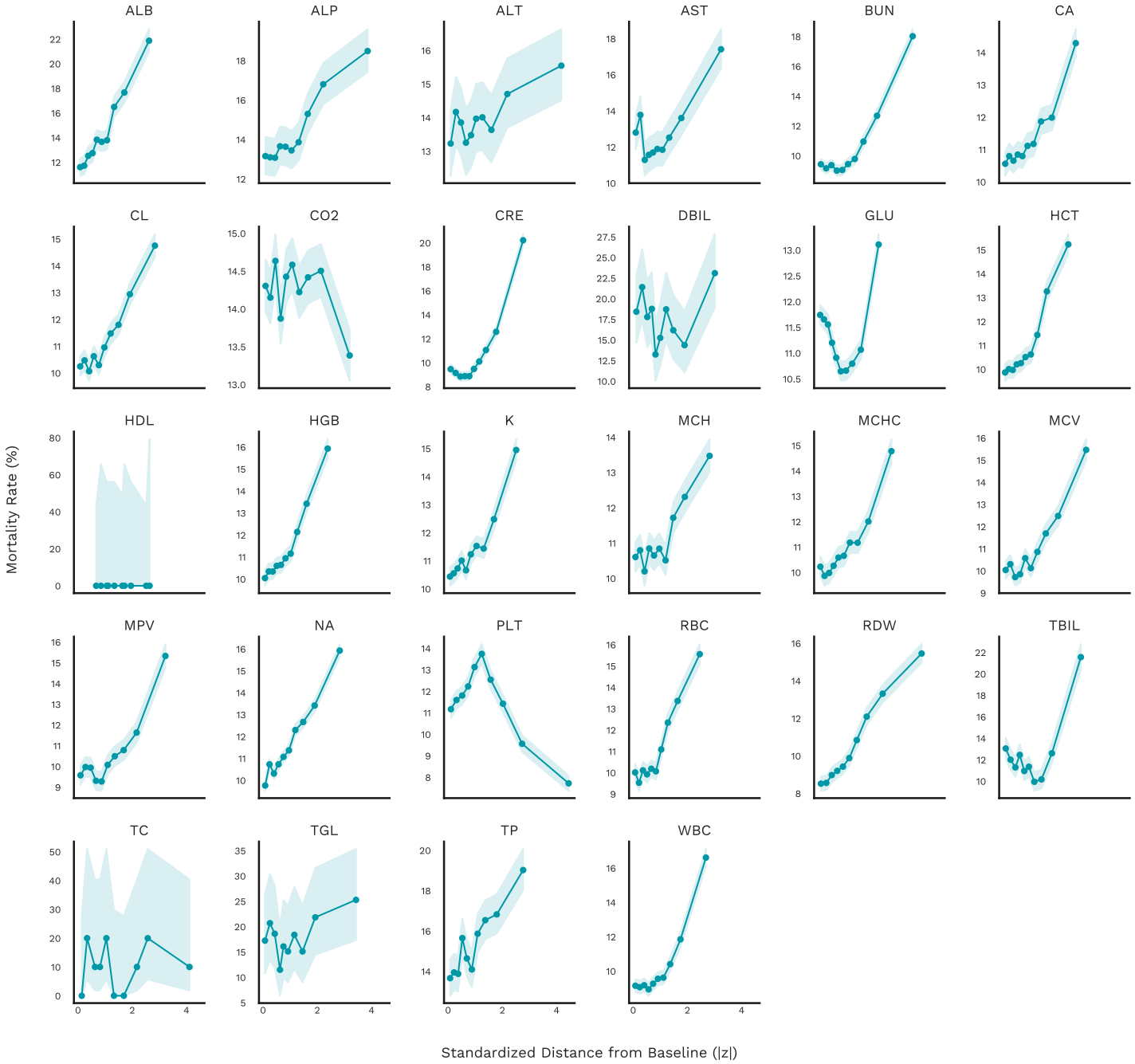


Figure S5: Supplementary Figure 5. Mortality rate by standardized deviation from personal baseline in eICU.

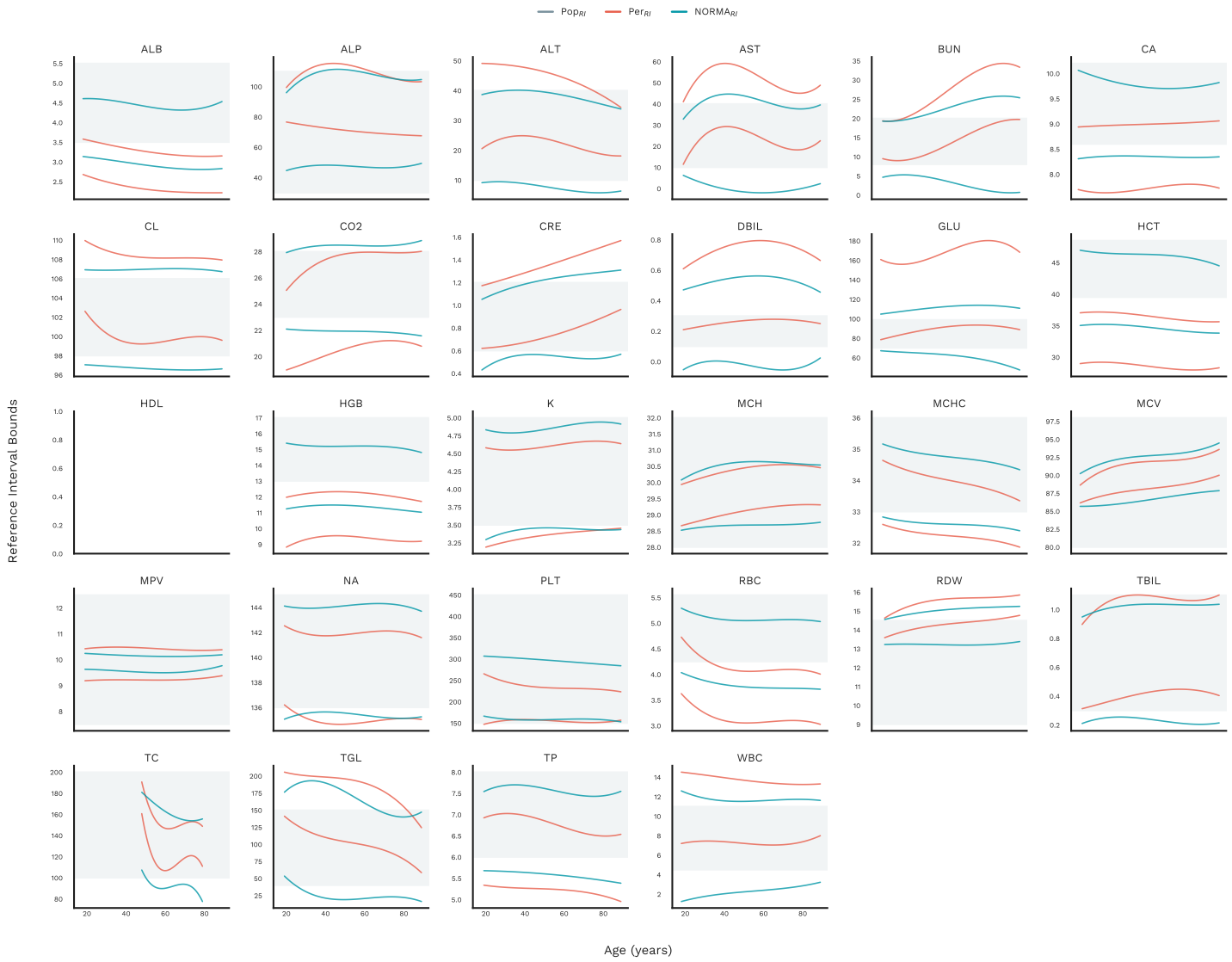


Figure S6: Supplementary Figure 6. Age-dependent reference interval trends in eICU.

3 Supplemental Tables

Analyte	Full Name	Unit	Population RI
A1C	Hemoglobin A1c	%	4.0–5.6
ALB	Albumin	g/dL	3.5–5.5
ALP	Alkaline Phosphatase	U/L	30–110
ALT	Alanine Aminotransferase	U/L	10.0–40.0
AST	Aspartate Aminotransferase	U/L	10.0–40.0
BUN	Blood Urea Nitrogen	mg/dL	8–20
CA	Calcium	mg/dL	8.6–10.2
CL	Chloride	mEq/L	98–106
CO2	Bicarbonate	mEq/L	23–28
CRE	Creatinine	mg/dL	F: 0.5–1.1, M: 0.7–1.3
DBIL	Direct Bilirubin	mg/dL	0.1–0.3
GLU	Glucose	mg/dL	70–99
HCT	Hematocrit	%	F: 37–47, M: 42–50
HDL	HDL Cholesterol	mg/dL	40–100
HGB	Hemoglobin	g/dL	F: 12.0–16.0, M: 14–18
K	Potassium	mEq/L	3.5–5.0
LDL	LDL Cholesterol	mg/dL	0–130
MCH	Mean Corpuscular Hemoglobin	pg	28–32
MCHC	MCH Concentration	g/dL	33–36
MCV	Mean Corpuscular Volume	fL	80–98
MPV	Mean Platelet Volume	fL	7.5–12.5
NA	Sodium	mEq/L	136–145
PLT	Platelet Count	$10^3/\mu$L	150–450
RBC	Red Blood Cell Count	$10^6/\mu$L	F: 4.0–5.2, M: 4.5–5.9
RDW	Red Cell Distribution Width	%	9–14.5
TBIL	Total Bilirubin	mg/dL	0.3–1.1
TC	Total Cholesterol	mg/dL	100–200
TGL	Triglycerides	mg/dL	40–150
TP	Total Protein	g/dL	6.0–8.0
WBC	White Blood Cell Count	$10^3/\mu$L	4.5–11.0

Table S1: Analyte Reference

	EHRSHOT		MIMIC-IV		eICU		CHS	
Patients	5,676		179,601		98,432		903,022	
Sequences	122,350		3,285,245		1,135,814		16,301,161	
Age	54.9 ± 17.2		58.2 ± 19.0		63.3 ± 16.1		—	
Sex	52% F / 48% M		54% F / 46% M		46% F / 54% M		—	
Time	2.5 [0.3, 7.2] yr		1.0 [0.1, 4.1] yr		6 [4, 10] days		—	
Tests/Person	33 [26, 34]		12 [8, 12]		15 [13, 16]		10 [9, 10]	
Analyte	N	Value	N	Value	N	Value	N	Value
A1C	1,745	6.3 ± 1.2	39,996	6.6 ± 1.3	—	—	336,494	7.0 ± 1.4
ALB	4,282	3.3 ± 0.8	55,220	3.8 ± 0.8	19,075	2.6 ± 0.7	790,490	4.3 ± 0.9
ALP	4,206	112.5 ± 58.8	78,930	107.0 ± 58.4	13,582	97.0 ± 43.2	1,099,044	75.1 ± 24.5
ALT	4,219	37.8 ± 25.2	92,599	30.2 ± 21.9	12,425	33.8 ± 23.4	1,233,902	20.9 ± 12.7
AST	4,220	31.9 ± 19.5	90,639	32.7 ± 21.4	12,704	37.1 ± 25.6	1,236,781	21.7 ± 8.9
BUN	5,180	23.0 ± 15.2	153,329	22.2 ± 14.3	57,540	25.3 ± 15.5	1,251,453	15.2 ± 5.2
CA	5,162	8.8 ± 0.7	121,478	8.8 ± 0.7	57,067	8.4 ± 0.7	909,036	9.4 ± 0.4
CL	5,172	102.4 ± 5.0	148,304	101.8 ± 5.2	59,172	103.8 ± 6.2	31,171	102.7 ± 3.1
CO2	5,187	25.9 ± 3.8	145,386	25.6 ± 4.1	70,208	25.0 ± 4.7	299	32.4 ± 13.8
CRE	5,161	1.1 ± 0.5	156,044	1.1 ± 0.6	54,820	1.1 ± 0.6	1,310,465	0.8 ± 0.2
DBIL	1,429	0.3 ± 0.3	6,635	0.7 ± 0.9	1,147	0.6 ± 0.5	—	—
GLU	5,150	124.9 ± 39.9	147,195	118.3 ± 40.4	90,958	139.6 ± 44.3	1,351,168	101.2 ± 28.6
HCT	5,470	32.2 ± 6.8	164,903	33.6 ± 6.8	60,110	30.1 ± 5.9	1,407,912	40.6 ± 4.0
HDL	1,549	53.6 ± 17.5	49,011	55.8 ± 17.7	11	26.0 ± 14.3	1,002,204	50.4 ± 13.3
HGB	5,369	10.7 ± 2.3	162,168	11.0 ± 2.3	60,482	9.9 ± 2.0	1,408,118	13.4 ± 1.4
K	5,236	4.1 ± 0.5	150,333	4.2 ± 0.5	66,569	4.0 ± 0.6	1,182,544	4.5 ± 0.4
LDL	821	94.9 ± 34.9	48,688	104.2 ± 37.8	3	92.1 ± 56.2	1,187,844	111.0 ± 31.8
MCH	5,294	30.4 ± 2.7	161,671	29.7 ± 2.8	48,432	29.7 ± 2.1	1,250,759	28.9 ± 2.1
MCHC	5,317	33.2 ± 1.4	161,806	32.6 ± 1.6	51,710	32.8 ± 1.3	1,407,568	33.1 ± 1.2
MCV	5,304	91.5 ± 6.9	161,661	91.2 ± 7.3	50,841	90.4 ± 5.7	1,400,648	87.1 ± 5.6
MPV	371	9.6 ± 1.6	—	—	37,083	9.7 ± 1.4	—	—
NA	5,193	137.4 ± 4.2	149,303	138.7 ± 4.2	62,419	138.6 ± 5.0	1,187,537	140.3 ± 2.6
PLT	5,320	198.0 ± 111.4	162,308	229.4 ± 113.3	53,371	199.0 ± 98.4	1,407,716	251.1 ± 64.3
RBC	5,312	3.5 ± 0.8	161,825	3.7 ± 0.8	53,584	3.4 ± 0.7	1,404,794	4.7 ± 0.5
RDW	5,315	16.1 ± 2.9	161,640	15.1 ± 2.4	48,715	15.8 ± 2.1	1,403,339	13.6 ± 1.2
TBIL	4,169	0.6 ± 0.4	73,002	0.7 ± 0.6	12,738	0.7 ± 0.5	815,774	0.6 ± 0.3
TC	1,635	173.5 ± 45.0	49,762	187.5 ± 44.6	35	108.3 ± 44.0	1,260,730	188.9 ± 37.9
TGL	4	165.7 ± 100.8	48,445	140.4 ± 77.5	312	144.9 ± 68.0	1,244,518	134.0 ± 65.6
TP	4,213	6.6 ± 1.0	18,673	7.0 ± 0.7	15,134	5.8 ± 1.0	791,481	7.3 ± 0.5
WBC	5,345	7.8 ± 4.6	164,291	7.7 ± 4.0	52,141	10.6 ± 4.6	1,409,027	7.1 ± 2.2

Table S2: Cohort Summary

Component	Design Choice	Description
Time embedding	Time2Vec	Encodes days since first measurement using learned linear and sinusoidal functions. Captures long-term temporal trends and periodic patterns.
	Delta-t	Encodes log-compressed time difference between consecutive measurements. Represents irregular sampling.
Health state encoding	Binary	Encodes each value as within or outside the population reference range. Does not distinguish high vs. low.
	Ternary	Encodes whether the value is low, normal, or high relative to the reference range. Handles asymmetric risk patterns.
Age encoding	Raw values	Uses age directly as numeric input. Assumes linear effects.
	Binned	Discretizes age into groups (e.g., decades) and maps each bin to a learned embedding. Captures nonlinear and threshold effects.
Lab sequence encoding	Raw values	Uses observed biomarker values directly. Preserves absolute clinical scale and keeps preprocessing minimal.
	Within-sequence normalization	Per-sequence z-scoring with learnable scale and shift, then denormalized at output. Stabilizes optimization and improves cross-analyte sharing.
Training objective	Gaussian NLL	Predicts mean and log-variance under a normal distribution. Aligns with conventional RI assumptions. May underperform for skewed or heavy-tailed labs.
	Quantile (pinball) loss	Predicts fixed quantiles (2.5, 25, 50, 75, 97.5%). Distribution-free, robust to skewness and heteroscedasticity.

Metric		NORMA Gaussian	NORMA Quantile	ARIMA	Mean	Last
MAE	Train	6.0 [3.2, 11.1]	5.7 [2.9, 10.1]	6.8 [2.5, 10.2]	8.4 [4.5, 15.0]	7.4 [3.7, 11.0]
	Val	6.0 [2.6, 10.7]	5.8 [2.9, 8.7]	6.8 [3.6, 10.0]	8.3 [4.9, 13.5]	7.3 [3.8, 10.3]
	Test	6.0 [2.6, 10.2]	5.9 [3.6, 9.1]	6.7 [4.3, 11.9]	8.4 [4.5, 13.6]	7.3 [3.9, 10.7]
MAPE	Train	10.9 [8.8, 14.0]	11.9 [8.6, 14.7]	15.9 [11.3, 20.1]	19.4 [14.8, 25.4]	14.8 [10.9, 19.6]
	Val	11.1 [7.9, 13.2]	12.6 [8.7, 15.6]	15.7 [12.1, 21.5]	19.7 [14.0, 27.4]	15.2 [10.6, 20.1]
	Test	11.1 [8.5, 14.3]	12.3 [7.9, 17.3]	16.9 [11.3, 23.3]	19.7 [13.5, 26.7]	15.2 [10.6, 21.1]
R2	Train	0.76 [0.72, 0.80]	0.70 [0.66, 0.75]	0.62 [0.57, 0.70]	0.46 [0.41, 0.49]	0.56 [0.47, 0.62]
	Val	0.75 [0.69, 0.78]	0.69 [0.66, 0.73]	0.62 [0.53, 0.67]	0.46 [0.43, 0.49]	0.55 [0.49, 0.62]
	Test	0.75 [0.70, 0.78]	0.68 [0.64, 0.72]	0.58 [0.51, 0.64]	0.45 [0.41, 0.50]	0.54 [0.49, 0.61]

Table S3: Prediction Performance

	NORMA Gaussian			NORMA Quantile			ARIMA		
Analyte	MAE	MAPE	R2	MAE	MAPE	R2	MAE	MAPE	R2
A1C	0.4	5.7	0.70	0.4	6.1	0.64	0.5	7.7	0.37
ALB	0.3	7.7	0.84	0.3	8.5	0.74	0.3	10.2	0.75
ALP	15.6	15.2	0.80	16.0	15.4	0.74	24.3	19.9	0.78
ALT	6.2	24.7	0.79	7.2	27.8	0.62	14.5	40.4	0.57
AST	6.0	21.0	0.77	7.0	23.5	0.58	20.4	44.5	0.41
BUN	3.3	17.2	0.83	3.3	18.4	0.83	4.4	23.2	0.82
CA	0.3	3.0	0.76	0.3	3.3	0.66	0.4	4.4	0.40
CL	1.8	1.8	0.75	1.9	1.8	0.69	2.4	2.3	0.55
CO2	1.3	5.4	0.80	1.5	6.3	0.70	2.1	8.3	0.48
CRE	0.1	12.3	0.83	0.1	10.6	0.88	0.2	12.9	0.87
DBIL	0.3	36.5	0.64	0.3	48.0	0.51	0.4	88.0	0.57
GLU	14.0	11.3	0.61	15.7	12.9	0.48	21.1	17.7	0.23
HCT	2.0	6.1	0.83	2.1	6.2	0.79	3.0	8.7	0.60
HDL	6.0	11.1	0.80	6.5	12.1	0.73	8.8	18.8	0.53
HGB	0.7	6.2	0.84	0.7	6.3	0.79	1.0	8.5	0.63
K	0.3	6.3	0.56	0.3	6.7	0.45	0.4	8.5	0.15
LDL	15.0	20.0	0.72	17.7	21.8	0.55	18.6	33.7	0.52
MCH	0.6	2.1	0.90	0.5	1.8	0.90	0.6	2.2	0.83
MCHC	0.6	1.9	0.72	0.6	1.8	0.72	0.6	1.9	0.58
MCV	1.9	2.2	0.85	1.6	1.8	0.88	1.7	1.9	0.82
MPV	0.7	6.7	0.66	0.6	6.5	0.59	0.9	9.9	0.21
NA	1.8	1.3	0.64	1.8	1.3	0.62	2.2	1.6	0.49
PLT	52.8	19.3	0.29	30.3	13.5	0.79	36.6	18.1	0.68
RBC	0.2	6.4	0.84	0.2	6.3	0.81	0.3	8.5	0.64
RDW	0.5	3.4	0.86	0.5	3.0	0.87	0.7	4.6	0.81
TBIL	0.1	22.5	0.81	0.2	28.0	0.71	0.3	32.1	0.85
TC	17.0	9.8	0.76	20.3	12.0	0.57	24.8	17.5	0.44
TGL	28.7	22.8	0.71	36.2	28.7	0.48	—	—	—
TP	0.3	4.7	0.74	0.4	5.7	0.54	0.5	6.9	0.57
WBC	1.3	19.6	0.79	1.5	24.2	0.62	2.1	26.0	0.69

Table S4: Analyte Performance

	CI Width Change (%)		q50 Shift (%)	
Feature	Median	IQR	Median	IQR
History Length	4.8	[1.0, 14.4]	0.0	[0.0, 0.0]
Prediction Horizon	1.6	[0.1, 11.1]	0.0	[0.0, 0.0]
Within-Person Variability	115.2	[111.4, 120.5]	0.7	[0.3, 1.7]

Table S5: Sensitivity Summary Norma Quantile

	History Length		Prediction Horizon		Within-Person Variability	
Analyte	CI (% ref)	q50 (%)	CI (% ref)	q50 (%)	CI (% ref)	q50 (%)
A1C	0.1 to 18.1	-0.0 to 0.0	0.1 to 16.7	-0.0 to 0.0	0.1 to 129.4	-0.0 to 0.6
ALB	0.1 to 14.5	-0.0 to 0.0	0.1 to 12.0	-0.0 to 0.0	0.1 to 118.6	-0.1 to 0.2
ALP	0.0 to 0.4	-0.0 to 0.0	0.0 to 0.0	-0.0 to 0.0	0.0 to 124.2	-0.0 to 2.1
ALT	0.0 to 1.0	-0.0 to 0.0	0.0 to 0.0	-0.0 to -0.0	0.0 to 127.7	-6.3 to -0.0
AST	0.0 to 1.0	-0.0 to 0.0	0.0 to 0.0	-0.0 to -0.0	0.0 to 125.8	-5.4 to -0.0
BUN	0.0 to 2.4	-0.0 to 0.0	0.0 to 1.2	-0.0 to 0.0	0.0 to 120.0	-0.0 to 0.8
CA	0.1 to 18.1	-0.0 to 0.0	0.1 to 8.8	-0.0 to 0.0	0.1 to 109.9	-0.0 to 0.4
CL	0.0 to 3.6	-0.0 to 0.0	0.0 to 1.6	-0.0 to 0.0	0.0 to 107.8	-0.0 to 0.1
CO2	0.0 to 5.8	-0.0 to 0.0	0.1 to 0.4	-0.0 to -0.0	0.1 to 110.1	-0.0 to 0.2
CRE	0.2 to 48.3	-0.0 to 0.1	0.3 to 34.2	-0.0 to 0.1	0.3 to 118.3	-0.0 to 0.5
DBIL	0.4 to 144.8	-0.1 to 0.7	0.7 to 46.5	-0.2 to -0.1	0.7 to 143.3	-0.1 to 2.7
GLU	0.0 to 1.0	-0.0 to 0.0	0.0 to 0.8	-0.0 to 0.0	0.0 to 113.5	-0.0 to 1.0
HCT	0.0 to 3.6	-0.0 to 0.0	0.0 to 2.2	-0.0 to 0.0	0.0 to 113.3	-0.1 to -0.0
HDL	0.0 to 0.5	-0.0 to 0.0	0.0 to 0.4	-0.0 to 0.0	0.0 to 120.6	-0.7 to 0.8
HGB	0.0 to 7.2	-0.0 to 0.0	0.1 to 5.9	-0.0 to 0.0	0.1 to 113.1	-0.4 to -0.0
K	0.1 to 19.3	-0.0 to 0.0	0.2 to 16.9	-0.0 to 0.0	10.5 to 113.7	0.0 to 0.7
LDL	0.0 to 0.2	-0.0 to 0.0	0.0 to 0.0	-0.0 to -0.0	0.0 to 116.4	-1.5 to 0.9
MCH	0.0 to 7.2	-0.0 to 0.0	0.0 to 6.4	-0.0 to 0.0	0.0 to 115.9	-0.0 to 0.2
MCHC	0.0 to 9.7	-0.0 to 0.0	0.1 to 7.8	-0.0 to 0.0	0.8 to 113.6	-0.3 to -0.0
MCV	0.0 to 1.6	-0.0 to 0.0	0.0 to 0.8	-0.0 to 0.0	0.0 to 115.7	0.0 to 0.8
MPV	0.0 to 5.8	-0.0 to 0.0	0.0 to 0.0	-0.0 to 0.0	0.0 to 44.6	-0.3 to 0.5
NA	0.0 to 3.2	-0.0 to 0.0	0.0 to 1.8	-0.0 to 0.0	0.2 to 110.7	-0.0 to 0.1
PLT	0.0 to 0.1	-0.0 to 0.0	0.0 to 0.0	-0.0 to -0.0	0.0 to 111.0	-4.0 to -0.0
RBC	0.1 to 20.7	-0.0 to 0.0	0.2 to 17.2	-0.0 to 0.0	0.2 to 115.6	-0.3 to -0.0
RDW	0.0 to 5.3	-0.0 to 0.0	0.0 to 0.5	-0.0 to -0.0	0.0 to 137.5	-0.0 to 0.8
TBIL	0.1 to 36.2	-0.0 to 0.2	0.2 to 17.2	-0.1 to 0.1	0.2 to 121.3	-4.3 to -0.0
TC	0.0 to 0.3	-0.0 to 0.0	0.0 to 0.0	-0.0 to -0.0	0.0 to 115.0	-0.2 to 0.4
TGL	0.0 to 0.3	-0.0 to 0.0	0.0 to 0.0	-0.0 to 0.0	0.0 to 114.5	-0.0 to 1.8
TP	0.1 to 14.5	-0.0 to 0.0	0.1 to 12.6	-0.0 to 0.0	0.1 to 112.9	-0.2 to 0.0
WBC	0.0 to 4.5	-0.0 to 0.0	0.0 to 1.7	-0.0 to 0.0	0.0 to 109.7	-1.4 to -0.0

Table S6: Sensitivity Norma Quantile

	CI Width Change (%)		μ Shift (%)	
Feature	Median	IQR	Median	IQR
History Length	28.3	[20.6, 41.8]	12.8	[5.6, 20.6]
Prediction Horizon	7.2	[3.0, 9.9]	1.6	[0.8, 3.2]
Within-Person Variability	7.2	[3.7, 12.4]	1.3	[0.3, 3.6]

Table S7: Sensitivity Summary Norma Gaussian

	History Length		Prediction Horizon		Within-Person Variability	
Analyte	CI (% ref)	μ (%)	CI (% ref)	μ (%)	CI (% ref)	μ (%)
A1C	53.2 to 80.6	-25.0 to 3.8	63.4 to 74.4	-1.5 to 0.6	63.4 to 66.8	0.5 to 3.2
ALB	50.3 to 60.9	-11.5 to -0.0	52.0 to 59.8	-2.8 to -0.5	52.0 to 61.7	-2.2 to -1.6
ALP	36.7 to 63.4	1.8 to 7.3	57.4 to 66.8	4.4 to 8.3	57.4 to 79.0	2.0 to 6.2
ALT	42.0 to 94.4	-29.6 to 2.9	71.9 to 85.5	-10.4 to 0.1	71.9 to 91.6	-3.6 to 0.1
AST	34.8 to 82.9	-27.8 to 3.7	63.8 to 81.3	-7.9 to 1.1	63.8 to 85.5	-2.5 to 1.1
BUN	42.9 to 90.8	-21.2 to 2.2	79.6 to 88.1	-1.8 to 1.5	80.3 to 88.3	-0.5 to 1.1
CA	78.2 to 106.2	-1.6 to 1.9	77.8 to 87.8	-0.1 to 0.5	79.3 to 84.2	0.1 to 0.6
CL	103.9 to 133.5	-2.1 to 0.7	103.7 to 104.1	0.2 to 0.3	104.1 to 104.5	0.3 to 0.4
CO2	89.4 to 116.9	-16.9 to 0.6	105.0 to 112.8	-2.2 to -0.7	106.4 to 108.8	-0.8 to -0.4
CRE	81.6 to 87.5	-6.8 to 8.5	80.0 to 83.0	-10.6 to 2.2	82.5 to 86.4	-6.8 to -6.3
DBIL	108.9 to 137.5	-44.1 to -6.6	126.4 to 135.2	-44.1 to -20.2	126.1 to 128.4	-44.4 to -33.0
GLU	42.7 to 103.7	1.0 to 7.7	97.0 to 99.9	1.3 to 2.1	97.5 to 105.6	0.0 to 1.6
HCT	87.3 to 152.6	-6.8 to 2.6	95.3 to 98.2	1.9 to 2.4	98.2 to 99.7	2.1 to 2.2
HDL	31.5 to 73.3	-2.1 to 14.9	68.9 to 75.7	-0.3 to 0.5	68.9 to 84.5	-1.7 to 1.8
HGB	58.7 to 79.1	-19.7 to -1.1	63.1 to 77.6	-3.2 to -1.8	63.1 to 69.1	-2.1 to -2.0
K	80.3 to 89.5	-1.5 to 2.5	80.0 to 84.4	-0.9 to 1.7	80.3 to 86.6	-0.8 to -0.5
LDL	33.9 to 55.1	-5.7 to 3.5	45.9 to 58.7	0.3 to 1.7	45.9 to 69.3	-0.9 to 8.2
MCH	66.8 to 105.4	-20.1 to 1.1	69.8 to 80.4	-0.4 to -0.0	71.4 to 78.2	-0.2 to 0.6
MCHC	72.9 to 114.8	-11.8 to 0.3	79.6 to 82.4	-0.3 to 0.1	80.7 to 84.4	0.0 to 0.2
MCV	57.3 to 69.9	-1.3 to 2.0	65.3 to 68.4	0.3 to 1.4	65.4 to 80.2	-0.1 to 0.6
MPV	12.2 to 31.8	-2.2 to 11.2	13.5 to 14.9	5.2 to 7.5	14.1 to 15.0	5.2 to 7.0
NA	94.9 to 101.7	0.1 to 0.7	94.9 to 95.9	0.1 to 0.2	94.9 to 95.6	0.1 to 0.2
PLT	85.4 to 109.4	-24.4 to -20.4	100.5 to 104.5	-20.5 to -20.5	102.2 to 122.2	-20.7 to -20.5
RBC	62.0 to 92.9	-5.9 to -1.1	62.0 to 69.7	-3.1 to -1.6	62.0 to 68.5	-2.7 to -2.6
RDW	31.9 to 43.2	-8.2 to 3.7	31.7 to 37.6	1.1 to 2.8	32.4 to 41.2	2.8 to 5.2
TBIL	36.1 to 103.9	-50.9 to -8.9	76.7 to 99.4	-33.1 to -18.2	78.4 to 84.8	-32.2 to -25.8
TC	51.5 to 80.2	-5.6 to 0.2	72.4 to 75.4	-1.4 to 0.3	73.4 to 80.9	-2.4 to -0.8
TGL	1.8 to 85.6	-35.2 to 17.4	75.4 to 80.1	4.3 to 6.3	75.5 to 88.2	-2.7 to 6.2
TP	67.8 to 101.2	-3.6 to 11.9	68.9 to 75.0	-0.1 to 1.1	70.7 to 82.1	0.2 to 1.2
WBC	56.2 to 82.9	-9.9 to 3.5	68.2 to 76.2	-1.0 to 3.9	69.0 to 79.5	-2.4 to 3.5

Table S8: Sensitivity Norma Gaussian

	eICU				CHS			
Analyte	N	CV_{intra}	CV_{inter}	II	N	CV_{intra}	CV_{inter}	II
A1C	—	—	—	—	336,494	0.1 [0.1, 0.1]	0.2 [0.2, 0.2]	0.4 [0.4, 0.4]
ALB	19,075	0.1 [0.1, 0.1]	0.2 [0.2, 0.2]	0.7 [0.7, 0.7]	790,490	0.1 [0.1, 0.1]	0.2 [0.2, 0.2]	0.7 [0.7, 0.7]
ALP	13,582	0.2 [0.2, 0.2]	0.4 [0.4, 0.4]	0.6 [0.5, 0.6]	1,099,044	0.1 [0.1, 0.1]	0.3 [0.3, 0.3]	0.4 [0.3, 0.4]
ALT	12,425	0.3 [0.3, 0.3]	0.6 [0.6, 0.6]	0.6 [0.6, 0.6]	1,233,902	0.2 [0.2, 0.3]	0.5 [0.5, 0.5]	0.5 [0.5, 0.5]
AST	12,704	0.4 [0.4, 0.4]	0.5 [0.5, 0.5]	0.7 [0.7, 0.7]	1,236,781	0.2 [0.2, 0.2]	0.2 [0.2, 0.2]	0.8 [0.8, 0.8]
BUN	57,540	0.3 [0.3, 0.3]	0.5 [0.5, 0.5]	0.6 [0.6, 0.6]	1,251,453	0.2 [0.2, 0.2]	0.3 [0.3, 0.3]	0.5 [0.5, 0.5]
CA	57,067	0.1 [0.1, 0.1]	0.1 [0.1, 0.1]	0.8 [0.8, 0.8]	909,036	0.0 [0.0, 0.0]	0.1 [0.0, 0.1]	0.8 [0.8, 0.8]
CL	59,172	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.7 [0.7, 0.7]	31,171	0.0 [0.0, 0.0]	0.1 [0.0, 0.1]	0.6 [0.6, 0.6]
CO2	70,208	0.1 [0.1, 0.1]	0.1 [0.1, 0.2]	0.8 [0.8, 0.8]	299	0.1 [0.1, 0.1]	0.3 [0.3, 0.4]	0.3 [0.3, 0.3]
CRE	54,820	0.2 [0.2, 0.2]	0.5 [0.5, 0.5]	0.4 [0.4, 0.4]	1,310,465	0.1 [0.1, 0.2]	0.2 [0.2, 0.2]	0.7 [0.7, 0.7]
DBIL	1,073	0.5 [0.4, 0.5]	0.7 [0.7, 0.7]	0.6 [0.6, 0.7]	815,774	0.1 [0.1, 0.2]	0.2 [0.2, 0.2]	0.8 [0.7, 0.8]
GLU	90,958	0.2 [0.2, 0.2]	0.2 [0.2, 0.2]	1.0 [1.0, 1.0]	1,351,168	0.1 [0.1, 0.1]	0.2 [0.2, 0.2]	0.3 [0.3, 0.4]
HCT	60,110	0.1 [0.1, 0.1]	0.2 [0.2, 0.2]	0.7 [0.7, 0.7]	1,407,912	0.1 [0.0, 0.1]	0.1 [0.1, 0.1]	0.5 [0.5, 0.5]
HDL	11	0.2 [0.2, 0.3]	0.5 [0.3, 0.6]	0.5 [0.3, 1.1]	1,002,204	0.1 [0.1, 0.1]	0.3 [0.3, 0.3]	0.4 [0.4, 0.4]
HGB	60,482	0.1 [0.1, 0.1]	0.2 [0.2, 0.2]	0.7 [0.7, 0.7]	1,408,118	0.1 [0.0, 0.1]	0.1 [0.1, 0.1]	0.5 [0.4, 0.5]
K	66,569	0.1 [0.1, 0.1]	0.1 [0.1, 0.1]	1.1 [1.1, 1.1]	1,182,544	0.1 [0.1, 0.1]	0.1 [0.1, 0.1]	0.9 [0.9, 0.9]
LDL	—	—	—	—	1,187,844	0.1 [0.1, 0.2]	0.3 [0.3, 0.3]	0.5 [0.5, 0.5]
MCH	48,432	0.0 [0.0, 0.0]	0.1 [0.1, 0.1]	0.3 [0.3, 0.3]	1,250,759	0.0 [0.0, 0.0]	0.1 [0.1, 0.1]	0.2 [0.2, 0.2]
MCHC	51,710	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.6 [0.6, 0.6]	1,407,568	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.9 [0.8, 0.9]
MCV	50,841	0.0 [0.0, 0.0]	0.1 [0.1, 0.1]	0.3 [0.3, 0.3]	1,400,648	0.0 [0.0, 0.0]	0.1 [0.1, 0.1]	0.2 [0.2, 0.3]
MPV	37,083	0.0 [0.0, 0.0]	0.1 [0.1, 0.1]	0.3 [0.3, 0.3]	1,407,716	0.1 [0.1, 0.1]	0.1 [0.1, 0.1]	0.6 [0.6, 0.6]
NA	124,270	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	0.8 [0.8, 0.8]	1,187,537	0.0 [0.0, 0.0]	0.0 [0.0, 0.0]	1.0 [1.0, 1.0]
PLT	53,371	0.2 [0.2, 0.2]	0.4 [0.4, 0.4]	0.6 [0.6, 0.6]	1,407,716	0.1 [0.1, 0.2]	0.3 [0.3, 0.3]	0.6 [0.6, 0.6]
RBC	53,584	0.1 [0.1, 0.1]	0.2 [0.2, 0.2]	0.7 [0.7, 0.7]	1,404,794	0.1 [0.0, 0.1]	0.1 [0.1, 0.1]	0.5 [0.4, 0.5]
RDW	48,715	0.0 [0.0, 0.0]	0.1 [0.1, 0.1]	0.3 [0.3, 0.3]	1,403,339	0.1 [0.1, 0.1]	0.1 [0.1, 0.1]	0.7 [0.7, 0.7]
TBIL	12,738	0.3 [0.3, 0.3]	0.5 [0.5, 0.5]	0.6 [0.6, 0.6]	815,774	0.1 [0.1, 0.2]	0.2 [0.2, 0.2]	0.8 [0.7, 0.8]
TC	35	0.2 [0.1, 0.2]	0.3 [0.2, 0.4]	0.5 [0.4, 0.8]	1,260,730	0.1 [0.1, 0.1]	0.1 [0.1, 0.1]	0.9 [0.9, 0.9]
TGL	312	0.3 [0.3, 0.3]	0.4 [0.4, 0.4]	0.7 [0.7, 0.8]	1,244,518	0.2 [0.2, 0.2]	0.2 [0.2, 0.3]	0.8 [0.8, 0.8]
TP	15,134	0.1 [0.1, 0.1]	0.1 [0.1, 0.1]	0.8 [0.8, 0.8]	791,481	0.1 [0.0, 0.1]	0.1 [0.1, 0.1]	0.7 [0.7, 0.7]
WBC	52,141	0.3 [0.3, 0.3]	0.3 [0.3, 0.3]	0.8 [0.8, 0.8]	1,409,027	0.2 [0.2, 0.2]	0.3 [0.3, 0.3]	0.6 [0.6, 0.6]

Table S9: Variability

Dataset	N	Pop_{RI} (%)	Per_{RI} (%)	NORMA_{RI} (%)	Per_{RI} RR (%)	NORMA_{RI} RR (%)
eICU	29	50.2	34.9	45.7	37.3	9.1
CHS	30	6.8	29.3	18.0	21.3	6.6

Table S10: Prevalence

		Prec			Sens			Spec			
Dataset	Outcome	Per_{RI}	NORMA_{RI}	Δ	Per_{RI}	NORMA_{RI}	Δ	Per_{RI}	NORMA_{RI}	Δ	Per_{R}
eICU	Acute Kidney Injury	0.15	0.16	+0.01	0.41	0.21	-0.19	0.58	0.80	+0.21	0.5
	Mortality	0.10	0.13	+0.03	0.44	0.26	-0.18	0.59	0.80	+0.21	0.5
	Prolonged ICU Stay (>7d)	0.23	0.27	+0.04	0.38	0.25	-0.12	0.58	0.80	+0.22	0.5
	Sepsis	0.18	0.17	-0.00	0.41	0.20	-0.22	0.59	0.79	+0.21	0.5
CHS	Chronic Kidney Disease	0.09	0.13	+0.04	0.28	0.72	+0.45	0.57	0.83	+0.25	0.8
	Mortality	0.07	0.17	+0.10	0.27	0.73	+0.46	0.57	0.83	+0.26	0.6
	Type 2 Diabetes	0.08	0.15	+0.07	0.28	0.72	+0.44	0.57	0.83	+0.26	0.7

Table S11: Eval Summary

			Per_{RI}				NORMA_{RI}			
Analyte	N	Events	Prec	Sens	Spec	Acc	Prec	Sens	Spec	Acc
A1C	—	—	—	—	—	—	—	—	—	—
ALB	891	130	0.14	0.23	0.75	0.68	0.17	0.15	0.88	0.77
ALP	8,843	1,712	0.19	0.58	0.41	0.44	0.21	0.18	0.83	0.71
ALT	7,391	1,408	0.17	0.51	0.42	0.44	0.18	0.19	0.80	0.68
AST	8,035	1,384	0.17	0.50	0.47	0.48	0.16	0.14	0.85	0.73
BUN	19,881	1,164	0.05	0.38	0.53	0.52	0.05	0.12	0.87	0.83
CA	16,083	2,079	0.13	0.34	0.67	0.63	0.15	0.20	0.84	0.75
CL	26,608	3,839	0.13	0.34	0.62	0.58	0.16	0.14	0.88	0.77
CO2	29,891	3,601	0.09	0.23	0.68	0.63	0.11	0.16	0.83	0.75
CRE	27,847	2,062	0.07	0.23	0.74	0.70	0.05	0.06	0.89	0.83
DBIL	289	78	0.36	0.23	0.85	0.68	0.33	0.10	0.92	0.70
GLU	9,534	1,257	0.11	0.20	0.74	0.67	0.16	0.10	0.92	0.81
HCT	2,919	266	0.10	0.47	0.58	0.57	0.10	0.13	0.89	0.82
HDL	—	—	—	—	—	—	—	—	—	—
HGB	3,383	255	0.09	0.44	0.64	0.62	0.08	0.14	0.87	0.81
K	57,801	8,390	0.14	0.34	0.65	0.61	0.17	0.20	0.83	0.74
LDL	—	—	—	—	—	—	—	—	—	—
MCH	32,416	4,932	0.16	0.48	0.54	0.53	0.17	0.28	0.76	0.68
MCHC	22,846	3,339	0.14	0.37	0.61	0.58	0.14	0.15	0.84	0.74
MCV	43,792	6,744	0.15	0.56	0.43	0.45	0.16	0.22	0.78	0.70
MPV	34,479	5,045	0.15	0.53	0.47	0.48	0.15	0.88	0.13	0.24
NA	40,769	6,154	0.14	0.40	0.56	0.53	0.15	0.61	0.37	0.40
PLT	33,546	4,508	0.13	0.65	0.30	0.35	0.13	0.39	0.59	0.56
RBC	4,595	429	0.11	0.31	0.73	0.69	0.11	0.14	0.88	0.82
RDW	16,275	1,681	0.10	0.43	0.57	0.55	0.11	0.17	0.84	0.77
TBIL	8,985	1,660	0.19	0.31	0.70	0.63	0.20	0.13	0.88	0.74
TC	11	3	0.29	0.67	0.38	0.45	0.33	0.33	0.75	0.64
TGL	176	42	0.21	0.48	0.43	0.44	0.22	0.14	0.84	0.68
TP	4,859	864	0.19	0.36	0.67	0.61	0.21	0.17	0.87	0.74
WBC	25,827	3,491	0.12	0.39	0.57	0.55	0.14	0.13	0.87	0.77

Table S12: Acute Kidney Injury

			Per$_{RI}$				NORMA$_{RI}$			
Analyte	N	Events	Prec	Sens	Spec	Acc	Prec	Sens	Spec	Acc
A1C	—	—	—	—	—	—	—	—	—	—
ALB	891	49	0.05	0.20	0.75	0.72	0.08	0.18	0.88	0.84
ALP	8,843	1,135	0.13	0.60	0.42	0.44	0.16	0.21	0.83	0.75
ALT	7,391	909	0.12	0.56	0.43	0.45	0.12	0.20	0.80	0.73
AST	8,035	786	0.10	0.53	0.48	0.48	0.11	0.17	0.85	0.79
BUN	19,881	1,043	0.05	0.48	0.54	0.54	0.06	0.16	0.87	0.83
CA	16,083	1,199	0.08	0.34	0.67	0.65	0.08	0.17	0.83	0.78
CL	26,608	2,122	0.08	0.36	0.62	0.60	0.10	0.16	0.88	0.82
CO2	29,891	2,365	0.06	0.23	0.69	0.65	0.08	0.16	0.83	0.78
CRE	27,847	1,863	0.08	0.31	0.75	0.72	0.10	0.15	0.90	0.85
DBIL	289	49	0.06	0.06	0.80	0.68	0.04	0.02	0.90	0.75
GLU	9,534	688	0.04	0.15	0.74	0.70	0.08	0.10	0.91	0.85
HCT	2,919	250	0.10	0.49	0.58	0.57	0.13	0.18	0.89	0.83
HDL	—	—	—	—	—	—	—	—	—	—
HGB	3,383	260	0.09	0.42	0.64	0.62	0.12	0.20	0.87	0.82
K	57,801	5,617	0.11	0.38	0.66	0.63	0.12	0.21	0.83	0.77
LDL	—	—	—	—	—	—	—	—	—	—
MCH	32,416	3,229	0.11	0.49	0.54	0.53	0.12	0.29	0.76	0.71
MCHC	22,846	1,971	0.09	0.41	0.62	0.60	0.10	0.18	0.85	0.79
MCV	43,792	4,275	0.10	0.61	0.43	0.45	0.13	0.30	0.79	0.74
MPV	34,479	3,508	0.11	0.59	0.48	0.49	0.11	0.91	0.13	0.21
NA	40,769	3,781	0.09	0.44	0.57	0.55	0.10	0.65	0.37	0.40
PLT	33,546	2,550	0.07	0.65	0.31	0.33	0.07	0.37	0.59	0.57
RBC	4,595	350	0.10	0.35	0.73	0.70	0.10	0.16	0.88	0.83
RDW	16,275	874	0.06	0.49	0.57	0.57	0.06	0.17	0.84	0.80
TBIL	8,985	1,035	0.13	0.35	0.70	0.66	0.15	0.17	0.88	0.80
TC	11	2	0.29	1.00	0.44	0.55	0.67	1.00	0.89	0.91
TGL	176	34	0.20	0.56	0.46	0.48	0.22	0.18	0.85	0.72
TP	4,859	484	0.11	0.38	0.67	0.64	0.11	0.16	0.86	0.79
WBC	25,827	1,695	0.07	0.42	0.58	0.57	0.07	0.15	0.87	0.82

Table S13: Mortality

			Per_{RI}				NORMA_{RI}			
Analyte	N	Events	Prec	Sens	Spec	Acc	Prec	Sens	Spec	Acc
A1C	—	—	—	—	—	—	—	—	—	—
ALB	891	158	0.17	0.24	0.75	0.66	0.23	0.16	0.88	0.76
ALP	8,843	2,882	0.32	0.57	0.41	0.46	0.35	0.19	0.84	0.62
ALT	7,391	2,412	0.31	0.55	0.42	0.46	0.33	0.20	0.80	0.61
AST	8,035	2,561	0.30	0.49	0.46	0.47	0.31	0.14	0.85	0.62
BUN	19,881	3,352	0.15	0.41	0.53	0.51	0.19	0.15	0.87	0.75
CA	16,083	2,941	0.19	0.35	0.68	0.62	0.23	0.21	0.84	0.72
CL	26,608	5,167	0.15	0.29	0.60	0.54	0.23	0.15	0.88	0.74
CO2	29,891	5,495	0.12	0.20	0.67	0.58	0.23	0.21	0.84	0.73
CRE	27,847	5,026	0.19	0.27	0.75	0.66	0.19	0.11	0.90	0.76
DBIL	289	137	0.56	0.20	0.86	0.55	0.50	0.09	0.92	0.53
GLU	9,534	718	0.05	0.18	0.75	0.70	0.13	0.15	0.92	0.86
HCT	2,919	467	0.16	0.42	0.57	0.55	0.17	0.12	0.89	0.76
HDL	—	—	—	—	—	—	—	—	—	—
HGB	3,383	486	0.13	0.33	0.63	0.59	0.15	0.14	0.87	0.76
K	57,801	11,154	0.20	0.35	0.66	0.60	0.26	0.23	0.84	0.72
LDL	—	—	—	—	—	—	—	—	—	—
MCH	32,416	7,342	0.25	0.51	0.55	0.54	0.28	0.31	0.77	0.66
MCHC	22,846	4,454	0.18	0.35	0.60	0.56	0.19	0.15	0.84	0.71
MCV	43,792	9,765	0.22	0.57	0.43	0.46	0.27	0.27	0.80	0.68
MPV	34,479	7,956	0.24	0.55	0.48	0.49	0.24	0.90	0.14	0.31
NA	40,769	8,509	0.17	0.36	0.55	0.51	0.21	0.64	0.37	0.43
PLT	33,546	7,568	0.22	0.67	0.30	0.39	0.27	0.48	0.62	0.59
RBC	4,595	778	0.18	0.29	0.73	0.66	0.19	0.14	0.88	0.76
RDW	16,275	2,917	0.19	0.45	0.57	0.55	0.20	0.18	0.84	0.72
TBIL	8,985	2,983	0.35	0.32	0.71	0.58	0.35	0.13	0.88	0.63
TC	11	1	0.00	0.00	0.30	0.27	0.33	1.00	0.80	0.82
TGL	176	117	0.66	0.54	0.44	0.51	0.59	0.14	0.81	0.36
TP	4,859	1,495	0.33	0.36	0.67	0.58	0.37	0.17	0.87	0.65
WBC	25,827	5,040	0.17	0.38	0.57	0.53	0.20	0.13	0.87	0.73

Table S14: Prolonged ICU Stay (>7d)

			Per_{RI}				NORMA_{RI}			
Analyte	N	Events	Prec	Sens	Spec	Acc	Prec	Sens	Spec	Acc
A1C	—	—	—	—	—	—	—	—	—	—
ALB	891	87	0.08	0.20	0.75	0.70	0.13	0.16	0.88	0.81
ALP	8,843	2,106	0.23	0.57	0.41	0.45	0.23	0.16	0.83	0.67
ALT	7,391	1,925	0.25	0.54	0.42	0.45	0.25	0.19	0.80	0.64
AST	8,035	2,036	0.24	0.49	0.47	0.47	0.23	0.14	0.85	0.67
BUN	19,881	2,759	0.12	0.40	0.53	0.51	0.12	0.11	0.87	0.76
CA	16,083	2,034	0.13	0.34	0.67	0.63	0.14	0.19	0.83	0.75
CL	26,608	4,164	0.14	0.34	0.62	0.57	0.17	0.13	0.87	0.76
CO2	29,891	4,190	0.11	0.25	0.68	0.62	0.12	0.14	0.83	0.73
CRE	27,847	4,045	0.14	0.25	0.74	0.67	0.13	0.09	0.89	0.78
DBIL	289	72	0.32	0.22	0.84	0.69	0.33	0.11	0.93	0.72
GLU	9,534	1,557	0.13	0.20	0.74	0.65	0.18	0.10	0.91	0.78
HCT	2,919	332	0.11	0.40	0.57	0.55	0.12	0.12	0.89	0.80
HDL	—	—	—	—	—	—	—	—	—	—
HGB	3,383	363	0.10	0.36	0.63	0.60	0.12	0.15	0.87	0.79
K	57,801	9,268	0.16	0.34	0.65	0.60	0.17	0.18	0.83	0.73
LDL	—	—	—	—	—	—	—	—	—	—
MCH	32,416	5,623	0.18	0.47	0.53	0.52	0.19	0.27	0.76	0.67
MCHC	22,846	3,540	0.15	0.36	0.61	0.57	0.15	0.15	0.84	0.74
MCV	43,792	7,707	0.18	0.57	0.43	0.45	0.19	0.23	0.79	0.69
MPV	34,479	5,542	0.16	0.52	0.47	0.48	0.16	0.87	0.13	0.25
NA	40,769	6,847	0.15	0.40	0.56	0.53	0.17	0.62	0.37	0.41
PLT	33,546	5,676	0.16	0.67	0.31	0.37	0.17	0.41	0.59	0.56
RBC	4,595	594	0.13	0.27	0.73	0.67	0.10	0.09	0.88	0.78
RDW	16,275	2,064	0.12	0.41	0.57	0.55	0.12	0.16	0.84	0.75
TBIL	8,985	2,286	0.26	0.30	0.70	0.60	0.25	0.12	0.87	0.68
TC	11	2	0.29	1.00	0.44	0.55	0.00	0.00	0.67	0.55
TGL	176	56	0.35	0.61	0.48	0.52	0.30	0.14	0.84	0.62
TP	4,859	1,118	0.24	0.35	0.67	0.59	0.26	0.16	0.87	0.70
WBC	25,827	3,531	0.12	0.36	0.57	0.54	0.12	0.11	0.87	0.76

Table S15: Sepsis

Analyte	NORMA-only flags	Later Pop_{RI} flag	Median lead (h)	IQR (h)
A1C	—	—	—	—
ALB	458	119	26.5	22.8–54.0
ALP	2,722	374	35.2	23.7–73.1
ALT	2,840	432	24.9	23.4–51.5
AST	2,070	319	24.9	23.3–50.6
BUN	5,767	1,209	24.6	22.1–48.2
CA	7,879	1,508	24.9	19.0–51.5
CL	7,724	2,248	24.8	22.3–48.6
CO2	15,069	6,065	24.5	18.7–48.6
CRE	5,285	766	24.4	20.3–48.0
DBIL	61	2	46.2	35.1–57.2
GLU	5,325	3,510	6.8	3.5–18.2
HCT	887	227	23.4	13.5–29.0
HDL	0	0	—	—
HGB	1,038	226	23.7	16.7–29.0
K	17,062	4,193	24.8	18.4–50.4
LDL	0	0	—	—
MCH	14,778	1,360	33.3	23.3–72.5
MCHC	6,131	1,122	25.7	22.9–54.8
MCV	18,737	917	46.3	23.6–76.9
MPV	85,001	3,941	29.6	22.8–71.7
NA	208,205	55,713	26.4	20.7–68.7
PLT	35,602	3,266	43.0	23.7–72.2
RBC	1,192	256	24.3	20.8–47.7
RDW	4,586	367	26.3	23.5–53.5
TBIL	2,104	227	27.4	23.2–69.6
TC	8	2	122.4	98.4–146.3
TGL	43	5	37.8	24.2–47.1
TP	1,918	334	29.6	23.4–88.7
WBC	6,250	1,285	24.7	22.5–49.4
Overall	458,742	89,993	25.7	22.8–51.5

Table S16: Lead Time

			Per_{RI}				NORMA_{RI}			
Analyte	N	Events	Prec	Sens	Spec	Acc	Prec	Sens	Spec	Acc
A1C	200,000	10,000	0.11	0.35	0.62	0.82	0.08	0.65	0.77	0.60
ALB	200,000	10,000	0.08	0.25	0.56	0.82	0.15	0.75	0.84	0.77
ALP	200,000	10,000	0.07	0.20	0.51	0.83	0.17	0.80	0.89	0.79
ALT	200,000	10,000	0.07	0.21	0.52	0.83	0.16	0.79	0.88	0.79
AST	200,000	10,000	0.08	0.25	0.56	0.81	0.16	0.75	0.84	0.78
BUN	200,000	10,000	0.06	0.16	0.49	0.82	0.20	0.84	0.91	0.83
CA	200,000	10,000	0.11	0.30	0.62	0.84	0.09	0.70	0.78	0.61
CL	200,000	10,000	0.10	0.33	0.61	0.82	0.09	0.68	0.79	0.65
CO2	200,000	10,000	0.10	0.33	0.61	0.82	0.09	0.67	0.79	0.66
CRE	200,000	10,000	0.05	0.15	0.50	0.80	0.23	0.86	0.90	0.85
DBIL	200,000	10,000	0.11	0.36	0.63	0.82	0.08	0.64	0.77	0.61
GLU	200,000	10,000	0.06	0.22	0.55	0.80	0.18	0.78	0.85	0.82
HCT	200,000	10,000	0.08	0.23	0.53	0.83	0.14	0.77	0.86	0.76
HDL	200,000	10,000	0.10	0.31	0.59	0.82	0.11	0.69	0.81	0.70
HGB	200,000	10,000	0.08	0.22	0.54	0.84	0.14	0.78	0.86	0.75
K	200,000	10,000	0.10	0.30	0.60	0.83	0.11	0.70	0.80	0.69
LDL	200,000	10,000	0.10	0.32	0.59	0.82	0.10	0.68	0.81	0.68
MCH	200,000	10,000	0.09	0.26	0.55	0.83	0.13	0.74	0.85	0.73
MCHC	200,000	10,000	0.10	0.28	0.61	0.84	0.10	0.72	0.79	0.66
MCV	200,000	10,000	0.09	0.27	0.57	0.83	0.11	0.73	0.82	0.70
MPV	200,000	10,000	0.11	0.38	0.64	0.82	0.07	0.62	0.76	0.58
NA	200,000	10,000	0.11	0.34	0.62	0.82	0.09	0.66	0.78	0.63
PLT	200,000	10,000	0.09	0.26	0.57	0.83	0.12	0.74	0.83	0.72
RBC	200,000	10,000	0.09	0.27	0.53	0.83	0.13	0.73	0.87	0.74
RDW	200,000	10,000	0.09	0.24	0.56	0.83	0.14	0.76	0.83	0.74
TBIL	200,000	10,000	0.09	0.28	0.58	0.82	0.13	0.72	0.82	0.74
TC	200,000	10,000	0.09	0.31	0.59	0.82	0.11	0.69	0.81	0.72
TGL	200,000	10,000	0.10	0.32	0.60	0.82	0.11	0.68	0.80	0.70
TP	200,000	10,000	0.09	0.28	0.57	0.82	0.14	0.72	0.82	0.75
WBC	200,000	10,000	0.09	0.29	0.58	0.82	0.12	0.71	0.82	0.73

Table S17: Chronic Kidney Disease

			Per_{RI}				NORMA_{RI}			
Analyte	N	Events	Prec	Sens	Spec	Acc	Prec	Sens	Spec	Acc
A1C	200,000	16,000	0.10	0.30	0.61	0.73	0.10	0.70	0.79	0.47
ALB	200,000	16,000	0.05	0.27	0.53	0.55	0.21	0.73	0.86	0.76
ALP	200,000	16,000	0.04	0.19	0.49	0.57	0.24	0.81	0.91	0.78
ALT	200,000	16,000	0.03	0.20	0.50	0.48	0.25	0.80	0.90	0.80
AST	200,000	16,000	0.05	0.25	0.54	0.53	0.23	0.75	0.86	0.77
BUN	200,000	16,000	0.07	0.36	0.52	0.55	0.18	0.64	0.88	0.73
CA	200,000	16,000	0.09	0.31	0.61	0.68	0.14	0.69	0.79	0.62
CL	200,000	16,000	0.10	0.22	0.60	0.77	0.11	0.78	0.80	0.46
CO2	200,000	16,000	0.09	0.38	0.60	0.65	0.12	0.62	0.80	0.61
CRE	200,000	16,000	0.06	0.29	0.51	0.54	0.20	0.71	0.89	0.76
DBIL	200,000	16,000	0.09	0.29	0.62	0.72	0.11	0.71	0.77	0.54
GLU	200,000	16,000	0.06	0.27	0.52	0.61	0.19	0.73	0.88	0.73
HCT	200,000	16,000	0.06	0.22	0.56	0.65	0.20	0.78	0.84	0.73
HDL	200,000	16,000	0.08	0.26	0.58	0.69	0.16	0.74	0.81	0.66
HGB	200,000	16,000	0.06	0.17	0.56	0.73	0.18	0.82	0.83	0.69
K	200,000	16,000	0.09	0.35	0.59	0.66	0.13	0.65	0.81	0.62
LDL	200,000	16,000	0.08	0.28	0.58	0.67	0.16	0.72	0.82	0.67
MCH	200,000	16,000	0.06	0.25	0.57	0.63	0.19	0.75	0.82	0.73
MCHC	200,000	16,000	0.08	0.15	0.60	0.79	0.15	0.85	0.80	0.60
MCV	200,000	16,000	0.08	0.26	0.59	0.71	0.14	0.74	0.81	0.62
MPV	200,000	16,000	0.11	0.35	0.62	0.72	0.08	0.65	0.77	0.37
NA	200,000	16,000	0.11	0.33	0.61	0.73	0.09	0.68	0.79	0.40
PLT	200,000	16,000	0.03	0.21	0.57	0.28	0.28	0.79	0.82	0.82
RBC	200,000	16,000	0.07	0.33	0.56	0.59	0.17	0.67	0.84	0.71
RDW	200,000	16,000	0.06	0.24	0.57	0.66	0.18	0.76	0.82	0.71
TBIL	200,000	16,000	0.05	0.28	0.55	0.49	0.23	0.72	0.85	0.78
TC	200,000	16,000	0.05	0.28	0.55	0.52	0.22	0.72	0.85	0.77
TGL	200,000	16,000	0.07	0.28	0.58	0.65	0.17	0.72	0.81	0.69
TP	200,000	16,000	0.07	0.28	0.56	0.65	0.17	0.72	0.83	0.69
WBC	200,000	16,000	0.07	0.30	0.58	0.64	0.16	0.70	0.82	0.69

Table S18: Mortality

			Per_{RI}				NORMA_{RI}			
Analyte	N	Events	Prec	Sens	Spec	Acc	Prec	Sens	Spec	Acc
A1C	200,000	12,000	0.03	0.16	0.47	0.68	0.25	0.84	0.93	0.84
ALB	200,000	12,000	0.06	0.26	0.54	0.73	0.18	0.74	0.86	0.79
ALP	200,000	12,000	0.06	0.21	0.50	0.74	0.20	0.79	0.90	0.80
ALT	200,000	12,000	0.05	0.22	0.51	0.71	0.21	0.78	0.89	0.81
AST	200,000	12,000	0.06	0.25	0.53	0.71	0.20	0.75	0.86	0.80
BUN	200,000	12,000	0.07	0.30	0.52	0.71	0.18	0.70	0.88	0.79
CA	200,000	12,000	0.10	0.33	0.62	0.79	0.09	0.67	0.78	0.59
CL	200,000	12,000	0.10	0.34	0.61	0.78	0.10	0.66	0.79	0.62
CO2	200,000	12,000	0.10	0.35	0.61	0.78	0.09	0.65	0.79	0.60
CRE	200,000	12,000	0.06	0.28	0.52	0.70	0.19	0.72	0.88	0.80
DBIL	200,000	12,000	0.11	0.38	0.63	0.78	0.08	0.62	0.77	0.54
GLU	200,000	12,000	0.02	0.15	0.46	0.51	0.30	0.86	0.94	0.87
HCT	200,000	12,000	0.07	0.24	0.55	0.76	0.17	0.76	0.85	0.76
HDL	200,000	12,000	0.09	0.32	0.59	0.77	0.12	0.68	0.81	0.68
HGB	200,000	12,000	0.07	0.23	0.56	0.77	0.17	0.77	0.84	0.76
K	200,000	12,000	0.09	0.31	0.60	0.78	0.11	0.69	0.80	0.66
LDL	200,000	12,000	0.09	0.33	0.59	0.76	0.12	0.68	0.81	0.69
MCH	200,000	12,000	0.08	0.20	0.56	0.81	0.16	0.80	0.84	0.73
MCHC	200,000	12,000	0.10	0.28	0.61	0.80	0.11	0.72	0.79	0.62
MCV	200,000	12,000	0.09	0.27	0.57	0.79	0.13	0.73	0.82	0.69
MPV	200,000	12,000	0.11	0.39	0.64	0.78	0.07	0.61	0.76	0.50
NA	200,000	12,000	0.11	0.36	0.62	0.78	0.09	0.64	0.78	0.57
PLT	200,000	12,000	0.08	0.26	0.57	0.77	0.16	0.74	0.83	0.74
RBC	200,000	12,000	0.08	0.28	0.55	0.76	0.15	0.72	0.85	0.74
RDW	200,000	12,000	0.07	0.25	0.56	0.76	0.16	0.75	0.83	0.75
TBIL	200,000	12,000	0.08	0.29	0.58	0.76	0.14	0.71	0.82	0.73
TC	200,000	12,000	0.09	0.31	0.59	0.76	0.14	0.69	0.81	0.72
TGL	200,000	12,000	0.09	0.33	0.60	0.76	0.13	0.68	0.80	0.70
TP	200,000	12,000	0.08	0.28	0.57	0.76	0.15	0.71	0.82	0.74
WBC	200,000	12,000	0.08	0.29	0.58	0.76	0.14	0.71	0.82	0.73

Table S19: Type 2 Diabetes